Jefferson Liu

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Education

University of Toronto — Toronto, ON

Honours BSc in Computer Science (Focus in AI) & Computational Cognitive Science — *September 2021 – May 2026*

Experience

AI/ML Research Engineer — Aug 2025 – May 2026

University of Toronto — *Toronto, ON*

- Built a knowledge-graph-centric **RAG** context engine on an agentic AI platform, using **Neo4j** graph storage with embeddings in **pgvector** and **PostgreSQL** backend, increasing factual grounding accuracy by **15%+** relative to base LLM
- Developed **LangChain** agents for agentic context optimization and a graph traversal search agent using the **Claude Agent SDK**; ran **prompt A/B testing** across **OpenAI** and **Anthropic** models to improve output quality.
- Implemented hybrid retrieval (vector + graph) with explainability and triplet rendering, improving retrieval precision by **25%** relative to vector only search; built an **MCP server** exposing knowledge-graph tools to LLM agents.

Fullstack Developer Intern — Sept 2024 – Aug 2025

Ministry of Public and Business Service Delivery and Procurement — *Toronto, ON*

- Modernized Transfer Payment Ontario, a payment platform used by hundreds of Ontario corporations, building **RESTful APIs** with **Java**/**Spring Boot** backends and **Angular**/**TypeScript** frontends.
- Built program assessment functionality for the Experience Ontario initiative, automating funding-program viability scoring from historical data using **Oracle SQL** and **Java**, with an **Angular**/**TypeScript** frontend.
- **Led development** of a web application for internal ministry program assessments used by Ontario Public Service employees, integrating client request handling, employee management, and administrative validation tools (**Java**, **Spring Boot**, **Angular**, **TypeScript**, **Oracle SQL**).

Software Developer Intern — May 2022 – Aug 2022

Ministry of Government and Consumer Services — *Toronto, ON*

- Built **AngularJS** and **React** frontends for internal web apps serving 100+ clients, meeting OPS Digital Accessibility (**AODA/WCAG**) compliance requirements.
- Developed internal administrative tools using **Java** and **XML**/**React** used by ministry managers

Projects

Encoder-Only Transformer with MLM Pretraining | *Python, PyTorch, PyTorch Lightning* — Jul 2026 – Aug 2026

- Implemented an encoder-only transformer with single-head **self-attention** in **PyTorch**, training it under a **masked language modeling** objective with custom `collate_fn`

Cross-Category Speech Corpus Analysis | *Python, pandas, NumPy, SciPy* — Mar 2026 – Apr 2026

- Built data pipeline in **pandas** over **10,000+ minutes** of transcribed YouTube speech across **20+ categories**, with custom **lemmatization**, tokenization, and category-stratified aggregation into time-partitioned datasets for research on LLM influence on spoken vocabulary
- Quantified per-word shift with smoothed **log-odds ratios** and **Woolf standard errors**, tested significance with two-proportion **z-tests** under **Bonferroni correction**, and ran **paired t-tests** across target/control pairs using **NumPy/SciPy**

Multi-Agent Repository Architecture Recovery | *Python, LangChain, LLM APIs* — Jan 2026 – Apr 2026

- Replicated **ArchAgent** (arXiv 2601.13007), implemented **multi-agent orchestration** to parse a production monorepo with **tree-sitter** for static dependency extraction, then applying adaptive file clustering and **context compression** to compress repository-scale code into LLM context windows.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, HTML/CSS, XML
AI / ML: RAG, LangChain, Claude Agent SDK, OpenAI Agents SDK, MCP, PyTorch, Keras, NumPy
Frameworks: React, Angular, FastAPI, Spring Boot, Node.js, FastMCP, FastAPI, Flask
Databases: PostgreSQL, pgvector, Neo4j, Oracle SQL, SQLite
Cloud & DevOps: AWS, Railway, Docker, Git, Linux/Unix CLI
Developer Tools: Cursor, Claude Code, GitHub Copilot, VS Code, IntelliJ, PyCharm